TypeScript Assignments – Arrays

Array Declaration

1. Create an array called marks that stores 5 numbers. Print all the elements.
2. Create an array called names that stores 5 strings. Print the first and last element.
3. Create an array of boolean values containing true, false, true, true.
4. Create a TypeScript array to store the names of 5 cities and print each city using a loop.
5. Create an array called prices that can store only numbers. Add 5 prices and calculate their total.

Readonly Arrays

6. Create a readonly array of 5 student names and print the names.
7. Create a ReadonlyArray containing 5 numbers.
8. Try to change the first element of a readonly array. Observe the TypeScript error.
9. Try to use push() on a readonly array. Explain why TypeScript doesn't allow it.
10. Create a readonly array of marks and calculate the highest mark without modifying the array.

Multidimensional Arrays

11. Create a 2D array containing the marks of 3 students in 3 subjects.
12. Create a 2D array representing a 3 × 3 matrix and print all elements.
13. Create a 2D array representing a classroom seating arrangement and print each seat.
14. Find the sum of all numbers in a 2D array.
15. Find and print the largest number from a 2D array.

push()

16. Create an array of 3 fruits and add one more fruit using push().
17. Add 3 numbers to an empty array using push().
18. Create an array of student names and add two new students using push().
19. Add multiple products to an existing product array using a single push().
20. Create an array of tasks. Add a new task using push() and print the updated array.

pop()

21. Create an array of 5 numbers and remove the last element using pop().
22. Create an array of fruits and remove the last fruit.
23. Store 5 student names and remove the last student.
24. Store 5 products and use pop() to remove the last product. Print the removed product.

25. Create an array of browser tabs. Remove the last opened tab using pop().

shift()

26. Create an array of 5 numbers and remove the first number.
27. Create an array of fruits and remove the first fruit using shift().
28. Create an array of student names and remove the first student.
29. Create a queue of 5 customers and remove the first customer using shift().
30. Create an array of tasks and remove the first task from the list using shift().

unshift()

31. Create an array of 3 fruits and add a new fruit at the beginning.
32. Add two new student names to the beginning of an existing array.
33. Create an array of numbers and add 100 as the first element.
34. Create a list of tasks and add a high-priority task at the beginning.
35. Create a list of messages and add a new message at the beginning of the array.

slice()

36. Create an array of 5 numbers and extract the first 3 elements using slice().
37. Create an array of 6 fruits and extract elements from index 2 to index 5.
38. Create an array of student names and create a new array containing the first 3 students without modifying the original array.
39. Given [10, 20, 30, 40, 50, 60], extract [30, 40, 50] using slice().
40. Create an array of 10 numbers and create a new array containing the last 4 elements using slice().

splice()

41. Create an array of 5 numbers and remove the third element using splice().
42. Create an array of fruits and replace one fruit with another using splice().
43. Create an array of student names and insert a new student at index 2.
44. Given [10, 20, 30, 40, 50], remove 30 and 40 using splice().
45. Create a product list and replace two existing products with two new products using splice().

map()

46. Given [1, 2, 3, 4, 5], create a new array containing double of every number.
47. Given [10, 20, 30, 40], create a new array containing each number increased by 5.
48. Given an array of names, create a new array containing the names in uppercase.
49. Given an array of prices, create a new array where each price has 18% GST added.

50. Given an array of student marks, create a new array containing "Pass" or "Fail" based on each mark.

filter()

51. Given [10, 15, 20, 25, 30], create a new array containing only even numbers.
52. Given an array of ages, create a new array containing only people aged 18 or above.
53. Given an array of marks, filter students who scored more than 60.
54. Given an array of product prices, filter products costing more than ■1,000.
55. Given an array of names, filter names whose length is greater than 5 characters.

reduce()

56. Find the sum of [10, 20, 30, 40, 50] using reduce().
57. Find the product of [1, 2, 3, 4, 5] using reduce().
58. Find the total price of an array containing product prices.
59. Find the highest number from an array using reduce().
60. Given an array of marks, calculate the average mark using reduce().

find()

61. Given [10, 20, 30, 40, 50], find the first number greater than 25.
62. Given an array of student names, find the first name starting with "A".
63. Given an array of marks, find the first mark greater than 80.
64. Given an array of products with id and name, find the product whose id is 103.
65. Given an array of employees with name and salary, find the first employee whose salary is greater than ■50,000.

every()

66. Check whether every number in [2, 4, 6, 8] is even.
67. Check whether every number in an array is greater than 10.
68. Check whether every student has scored at least 35.
69. Check whether every product price is greater than ■100.
70. Given an array of employees, check whether every employee has an age of 18 or above.

some()

71. Check whether an array contains at least one even number.
72. Check whether an array contains at least one number greater than 100.
73. Check whether at least one student scored more than 90.
74. Check whether at least one product costs more than ■50,000.

75. Given an array of employees, check whether at least one employee has the role "Manager".